

Blinkit Grocery Sales Dashboard

A Power BI Data Analytics Project

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1. Project Introduction:

This project focuses on analyzing Blinkit grocery sales data using Microsoft Power BI. An interactive dashboard was developed to understand sales performance, product category trends, outlet performance, and customer ratings.

2. Objective:

The main objective of this project is to transform raw grocery sales data into meaningful business insights through data cleaning, visualisation, and analytical reporting.

Objectives:

- Analyse overall sales performance
- Identify top-performing product categories
- Compare outlet performance
- Understand customer ratings
- Create an interactive business dashboard

3. Dataset Description:

The dataset contains grocery sales information related to items, outlets, sales performance, and customer ratings. It includes product-level details and outlet-level attributes that help analyse business performance.

The dataset consists of the following attributes:

Column Name	Description
Item Fat Content	Category of product based on fat content (Low Fat / Regular)
Item Identifier	Unique identifier assigned to each product item
Item Type	Category/type of grocery item
Outlet Establishment Year	Year in which the outlet was established
Outlet Identifier	Unique identifier assigned to each outlet
Outlet Location Type	Classification of outlet location (Tier 1, Tier 2, Tier 3)
Outlet Size	Size category of outlet (Small, Medium, High)
Outlet Type	Type/category of outlet
Item Visibility	Percentage visibility of an item within the outlet
Item Weight	Weight of individual grocery items
Sales	Revenue generated from item sales
Rating	Customer rating received for the item/outlet

4. Data Cleaning Process:

After importing the Blinkit grocery dataset into Power BI, data cleaning and transformation were performed using Power Query to ensure consistency and accuracy in the dataset.

During the cleaning process, the Item Fat Content column was identified as having inconsistent category values representing the same product classification. The values such as "low fat", "LF", and "Low Fat" were standardized into a single category named "Low Fat". Similarly, the values "reg" and "Regular" were standardized into the category "Regular" by using the Replace Values option in Power Query.

This transformation helped remove duplicate categories and ensured accurate +grouping and analysis while creating visualizations in the dashboard.

After standardizing the categorical values, the cleaned dataset was prepared for further analysis and dashboard development in Power BI.

5. Data Modelling and DAX Measures:

After completing data cleaning and transformation, calculated measures were created using Data Analysis Expressions (DAX) in Power BI. These measures were used to calculate key performance indicators (KPIs) required for the dashboard.

The following measures were created from the BlinkIT Grocery Data table:

5.1 Total Sales

Total Sales = SUM('BlinkIT Grocery Data'[Sales])

Purpose:

This measure calculates the total revenue generated from all grocery items by adding the sales values across the dataset. It is used to display the overall sales performance in the KPI card.

5.2 Average Rating

Average Rating = AVERAGE('BlinkIT Grocery Data'[Rating])

Purpose:

This measure calculates the average customer rating across all items, helping to understand overall customer satisfaction.

5.3 Average Sales

Average Sales = AVERAGE('BlinkIT Grocery Data'[Sales])

Purpose:

This measure calculates the average sales value per item and provides insight into the typical sales performance of products.

5.4 Number of Items

No of Items = COUNTROWS('BlinkIT Grocery Data')

Purpose:

This measure counts the total number of records/items available in the dataset and represents the total number of grocery items analysed.

6. Dashboard Development:

6.1 Dashboard Theme:

Used Blinkit-inspired:

- Yellow background elements
- Green highlights
- Rounded cards
- Interactive filters

6.2 KPI Section

Created cards for:

KPI	Value
Total Sales	1.20M
Average Sales	141
Average Rating	3.9
Number of Items	8523

7. Dashboard Visualizations Created

After creating the required DAX measures, different Power BI visuals were developed to analyze sales performance, product characteristics, and outlet-related factors. Each visualization was designed to provide meaningful business insights from the Blinkit grocery dataset.

7.1 Outlet Establishment Analysis:

Visualization Used: Line Chart

The line chart was created to analyze sales performance based on the Outlet Establishment Year. This visualization helps understand how sales have changed over different years of outlet establishment and identifies trends in outlet performance over time.

7.2 Fat Content Analysis:

Visualisation Used: Donut Chart

The donut chart was created to analyse the contribution of different Item Fat Content categories towards total sales. The categories were divided into Low Fat and Regular products to understand customer preference and sales distribution.

7.3 Item Type Analysis

Visualisation Used: Bar Chart

The bar chart was created to compare sales performance across different Item Types. This helps identify which grocery categories contribute more towards overall revenue.

7.4 Outlet Size Analysis

Visualisation Used: Donut Chart

The donut chart was used to analyse sales distribution based on Outlet Size categories. This visualisation helps compare the contribution of different outlet sizes to total sales.

7.5 Outlet Location Analysis

Visualisation Used: Funnel Chart

The funnel chart was created to compare sales performance across different Outlet Location Types such as Tier 1, Tier 2, and Tier 3 locations.

7.6 Outlet Type Analysis

Visualization Used: Matrix Table

The matrix visualization was created to provide a detailed comparison of different Outlet Types using key metrics such as:

- Total Sales
- Number of Items
- Average Sales
- Average Rating
- Item Visibility

This provides a detailed overview of outlet performance.

8. Dashboard Visualizations Created

Different Power BI visuals were created to analyze sales performance, product characteristics, and outlet performance.

Visualization	Chart Type	Purpose
Outlet Establishment Analysis	Line Chart	Analyses sales trends based on outlet establishment year.
Fat Content Analysis	Donut Chart	Compares sales contribution between Low Fat and Regular products.
Item Type Analysis	Bar Chart	Identifies sales performance across different item categories.
Outlet Size Analysis	Donut Chart	Shows sales distribution based on outlet size categories.
Outlet Location Analysis	Funnel Chart	Compares sales performance across different outlet location types.
Outlet Type Analysis	Matrix	Provides detailed comparison of outlet types using sales, items, ratings, and visibility metrics.

